HDOC Day-53

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **nums[]**. Pass the array to a function named **product_except_self()** using its address. Use pointers to pass the array between the calling function and **product_except_self()**.

Create an output array **answer[]** of the same size. For each index **i**, store the product of all elements of **nums[]** except **nums[i]**. The product must not include the current element and division must not be used. Display the result in the original order, separated by commas.

C Program

```
#include <stdio.h>

void product_except_self(int *nums, int size)
{
    int answer[size];
    int prefix = 1, suffix = 1;
    int i;

    /* Store product of all elements to the left */
    for (i = 0; i < size; i++)
    {
        *(answer + i) = prefix;
        prefix = prefix * (*(nums + i));
    }

    /* Multiply by product of all elements to the right */
    for (i = size - 1; i >= 0; i--)
    {
        *(answer + i) = (*(answer + i)) * suffix;
        suffix = suffix * (*(nums + i));
    }

    printf("Output: ");
    for (i = 0; i < size; i++)
    {
        printf("%d", *(answer + i));
        if (i < size - 1)
            printf(", ");
    }
    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", nums + i);

    product_except_self(nums, size);

    return 0;
}
```

Explanation

The array is passed to **product_except_self()** through a pointer. In the first loop, **answer[i]** stores the product of all elements to the left of **nums[i]**. In the second loop, that value is multiplied by the product of all elements to the right. Therefore, the final value contains the product of every array element except the element at the same index. No division is used.

Sample Outputs

```
Example 1
Array:  1 2 3 4
Output: 24, 12, 8, 6

Example 2
Array:  -1 1 0 -3 3
Output: 0, 0, 9, 0, 0
```

Result

Thus, the program successfully calculates the product of all array elements except the current element using pointers, prefix and suffix products, without using division.